

Peiyao Tao

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Computer Vision & AI Engineer | Focused on deep learning, data-driven optimization, and scalable perception systems.

Technical Skills

Computer Vision: OpenCV, YOLO, MATLAB, Transformers, CNN, Object Detection, Reflection Removal, VLA, VLM

AI/ML: HuggingFace, Scikit-learn, Onnx, LangChain, Pytorch, TensorFlow, MCP, Data Engineering

Languages: Python, C++, C, Java, C#, HTML, CSS, JavaScript

Cloud & Infrastructure: Google Cloud, AWS (EC2, Lambda), Azure, Docker

Domains: Computer Vision, Agentic AI, NLP, Distributed Systems

Professional Experience

Full Stack Developer

DA CHANG AI TECHNOLOGY LLC | Seattle, WA | 01/2026 – Present

- Architected a scalable multi-tenant backend utilizing subdomain routing for strict merchant data isolation and designed a nested menu data model with atomic transactions and historical pricing snapshots.
- Implemented a JWT-based Role-Based Access Control system to secure distinct Customer, Staff, and Manager portals, and developed an interactive drag-and-drop floor plan editor for real-time table state tracking.
- Resolved N+1 database querying and race conditions via session logic, cutting API latency by over 60%, and integrated automated physical ticket printing to streamline kitchen order fulfillment.

Full Stack Developer Intern

1928 Tap & Grill Catering Co. | Seattle, WA | 05/2025 – 08/2025

- Engineered a full-stack digital menu and inventory management system, including a normalized SQLite database.
- Developed a secure merchant portal for staff to manage menu items and inventory in real-time.
- Integrated the system via a RESTful API for seamless communication between customer menu, merchant portal, and database.

Computer Vision Engineer Intern

FengHua Railway Informatization Co., Ltd | Zhengzhou, China | 02/2024 – 08/2024

- Developed a YOLOv5-based object detection system for railway surveillance that accurately identifies *persons, bushes, and new tracks* from live video feeds, achieving 95% detection accuracy and real-time inference at >30 FPS.
- Created and augmented a proprietary dataset of 3,000+ labeled images through manual annotation and data augmentation (rotation, color jitter, blur) to improve model generalization under varied lighting and weather conditions.
- Fine-tuned pre-trained YOLOv5 COCO weights on the curated dataset using transfer learning, optimizing hyperparameters (learning rate, batch size) to boost mAP by 12% over the baseline model.
- Evaluated and deployed the final model to production pipelines, ensuring stable performance during continuous real-time monitoring of track environments.

Software Projects

Vision-Language-Action (VLA) Agent for MiniGrid (<https://github.com/PeiyaoTao/VLM-for-RL-in-Embodied-Perception>)

- Developed a neuro-symbolic VLA agent that solved sparse-reward navigation tasks with a ~90% success rate, utilizing a custom Spatial Softmax perception module and CLIP embeddings to overcome "physics blindness."
- Implemented a 4-phase curriculum learning strategy with dynamic reward shaping and transfer learning, training a Recurrent PPO (LSTM) policy to master multi-stage logic puzzles where standard baselines failed.

Reflection Removal Models (https://github.com/PeiyaoTao/Comparison_of_Reflection_Removal)

- Designed and benchmarked multiple reflection removal architectures, including a classical optimization-based model, a U-Net with ResNet50 encoder, and an advanced Laplacian Pyramid Transformer for layered image decomposition.
- Implemented a composite loss function integrating L1, perceptual (VGG), exclusion, and gradient terms, improving reconstruction fidelity and visual realism.
- Trained and fine-tuned models on synthetic VOC2012 data with data augmentation and evaluated on real-world SIR² test sets, achieving notable PSNR/SSIM gains and consistent qualitative improvements across lighting conditions.

Presentations & Slides Builder Agent (<https://github.com/PeiyaoTao/Presentations-Builder-Agent>)

- Designed a multi-stage LangGraph agent with tool selection, JSON validation, and self-repair logic to convert unstructured PDFs into presentations.
- Engineered a generative NLP pipeline with a map-reduce summarizer and JSON-enforcing outline generator, implementing custom heuristics to preserve quantitative data and ensure quality standards. Integrated with the PowerPoint MCP server to enhance the creation of PowerPoint slides.

Discord Study Group Chatbot

- Designed and deployed a Discord chatbot from scratch, enabling students to create, join, and manage study groups, featuring a customizable reminder system with personalized lead times, all supported by MongoDB for persistent, real-time data management.

Education

M.S. in Computer Science @ Northeastern University | 09/2023 - 12/2025

Courses: Computer Vision, Advanced Perception, Machine Learning, Natural Language Processing, Advanced Algorithms & Data Structures, Database, Computer Networking

B.S. in Computer Science (Major), Mathematics & Human-Computer Interaction(Minors) @ Virginia Tech | 09/2018 - 05/2023

Courses: Artificial Intelligence, Computer Systems, Numerical Analysis, Linear Algebra, Differential Equations, Multivariable Calculus, Applied Combinatorics, Human-Computer Interaction, GUI programming and Graphics